MILITARY STANDARD

TRUCKLOADING

ROCKET MOTER CLUSTER; LAU-10/A,LAU-10A/A
LAU-10B/A OR LAU-10C/A LAUNCHER (ZUNI)
FLEET ISSUE UNIT LOAD

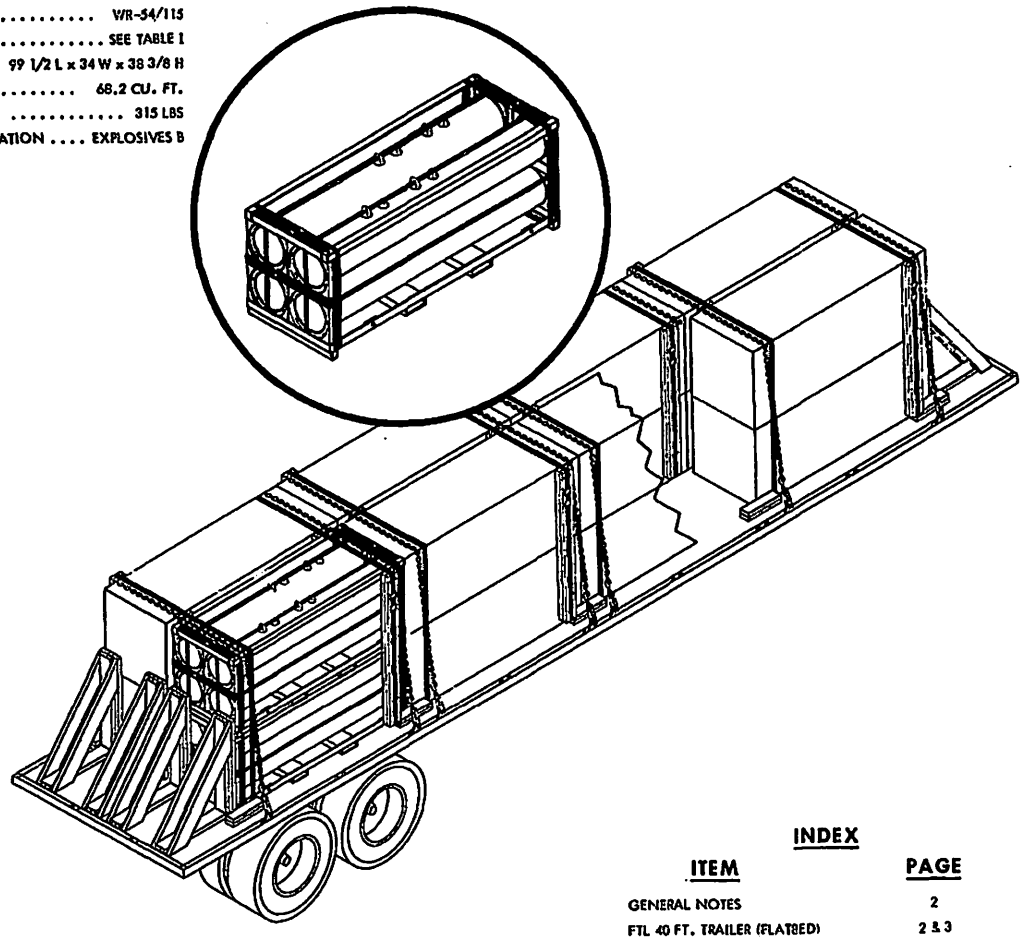
MIL-STD-1320-140A
(NAVY)

15 MAY 1974

SUPERSEDING
MIL-STD-1320-140
(NAVY)
2 JULY 1973

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/115
WEIGHT ..... SEE TABLE I
DIMENSIONS ..... 99 1/2 L x 34 W x 38 3/8 H
CUBE ..... 68.2 CU. FT.
WT. OF BOTTOM FRAME ..... 315 LBS
DOT HAZARD CLASSIFICATION .... EXPLOSIVES B



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LTL 40 FT. TRAILER (FLATBED)	4, 5 & 6
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NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

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THIS DOCUMENT CONTAINS 7 PAGES.

M.S. Gray NWHL 5/1/74
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 5/1/74
SIGNATURE AIRSYSCOM BY DIRECTION DATE

ORIGINATOR Charles McDonald
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING FLEET ISSUE UNIT LOADS OF LAU-10/A, LAU-10A/A, LAU-10B/A OR LAU-10C/A LAUNCHERS. THE LAUNCHERS SHALL BE PALLETIZED IN ACCORDANCE WITH WR-54/115.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS WITH THE TRAILERS TANDEM AXLES LOCATED IN THE "WESTERN POSITION" (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. ALL TWO HIGH UNIT LOADS SHALL BE UNITIZED AS SHOWN BELOW IN THE UNITIZING PROCEDURE.

WARNING

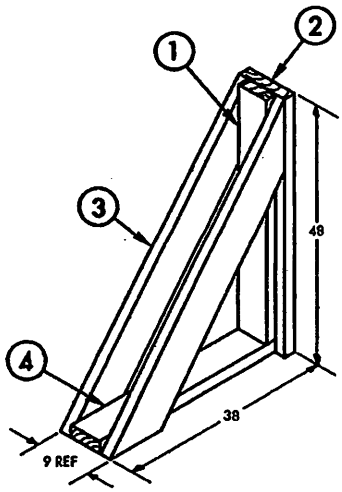
THE STACKING AND STRAPPING TOGETHER OF UNIT LOADS IS AUTHORIZED AS PART OF THIS TRUCKLOADING PROCEDURE ONLY. IT IS INTENDED TO ENSURE CONTINUOUS ENGAGEMENT OF THE STACKING FEATURES AND PROVIDE A GOOD TRUCKLOAD. MOVEMENT OF UNIT LOADS IN THIS CONFIGURATION SHALL BE LIMITED TO LOADING AND UNLOADING THE TRAILER.

4. KNEE BRACE, DETAIL A (OR D) SHOULD BEAR FIRM AGAINST THE UPPER AND LOWER UNIT LOAD (OR SPACER ASSEMBLY). SHIM IF REQUIRED TO ELIMINATE VOID. SECURE SHIMS BY NAILING SHIM TO KNEE BRACE
5. CHAINS AND LOADBINDERS SHALL BE USED TO SECURE THE UNIT LOADS TO THE TRAILER AND THEY SHALL MEET THE FOLLOWING CONDITIONS:
   a. CHAINS AND FITTINGS SHALL BE 3/8 INCH AND LOADBINDERS 3/8 INCH CAPACITY.
   b. CHAINS, FITTINGS AND LOADBINDERS SHALL HAVE A MINIMUM SAFE WORKING LOAD OF 5000 LBS.
   c. LOADBINDERS SHALL BE SAFETY WIRED WITH 16 GAUGE SOFT ANNEALED IRON WIRE OR SECURED USING THE END PIECE OF THE 3/8 CHAIN.
6. PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOADBINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOADBINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOADBINDER.
7. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
8. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO TRAILER IN AN ACCESSIBLE AREA AND ATTACH "EXPLOSIVES 8" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
9. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
10. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

TABLE I

DODIC / NALC	ITEM DESCRIPTION	UNIT LOAD WT (LBS) APPROX.	TRUCKLOAD WT (LBS) APPROX.
H546	ROCKET MOTOR CLUSTER, 5 INCH ROCKET MOTOR MK 16 & MODS, W/PROPELLANT GRAIN MK 49 MODS, 4 MOTORS PER LAU-10/A OR LAU-10A/A LAUNCHER.	2,014	32,224
H547	ROCKET MOTOR CLUSTER, 5 INCH ROCKET MOTOR MK 70 MOD 0, W/PROPELLANT GRAIN MK 49 MODS, 4 MOTORS PER LAU-10B/A LAUNCHER.	2,050	32,800
H548	ROCKET MOTOR CLUSTER, 5 INCH ROCKET MOTOR MK 71 MOD 1, W/PROPELLANT GRAIN MK 85 MOD 0, 4 MOTORS PER LAU-10C/A LAUNCHER.	2,254	30,064

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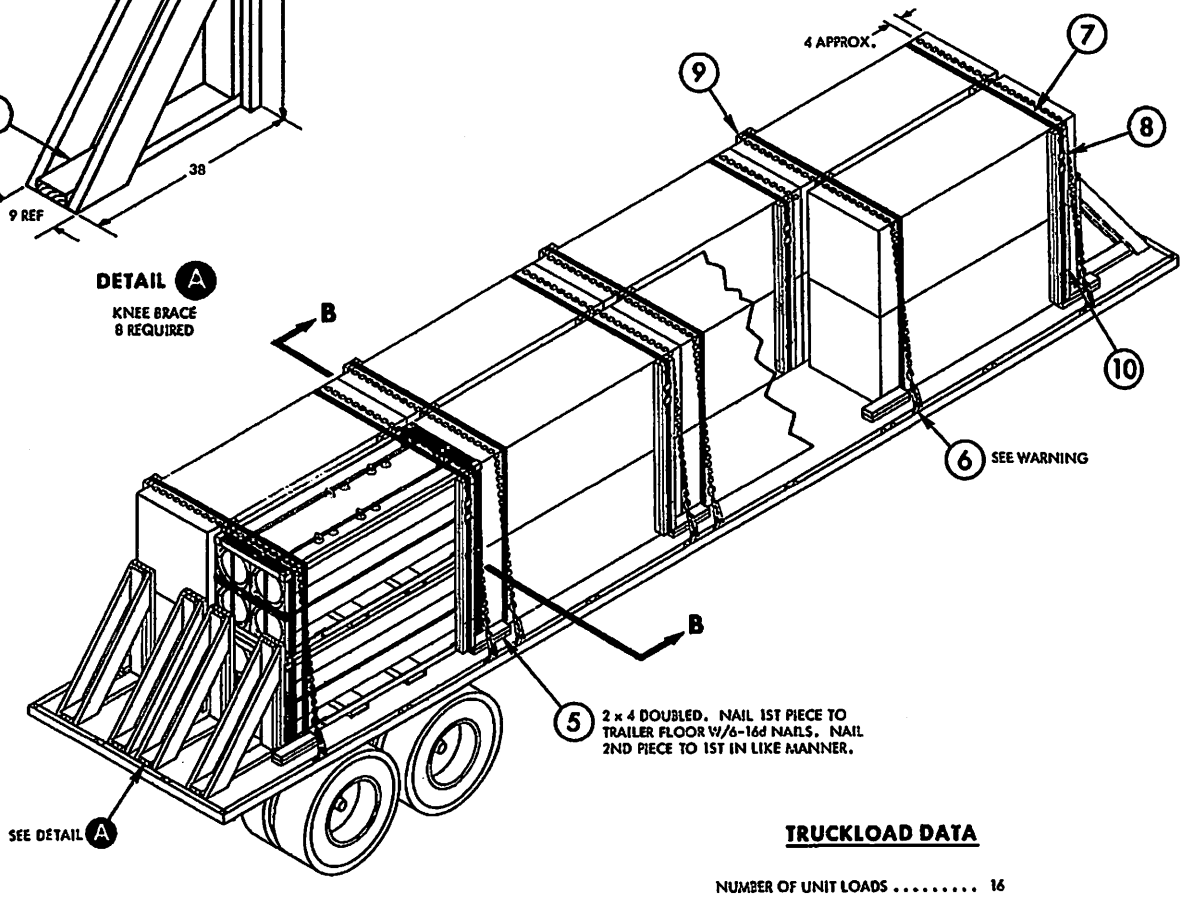


DETAIL A
KNEE BRACE
8 REQUIRED

WARNING

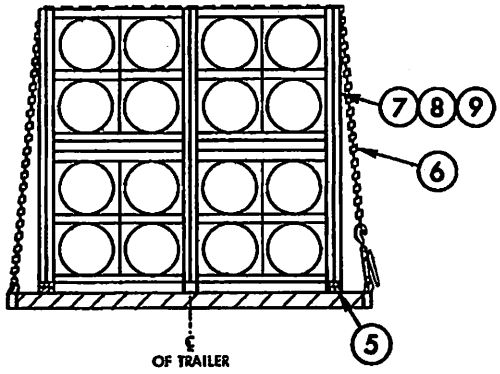
DO NOT POSITION CHAIN OVER STEEL STRAPPING.

POSITION FORWARD UNIT LOADS 41 INCHES FROM FRONT OF TRAILER.



TRUCKLOAD DATA

NUMBER OF UNIT LOADS 16
WEIGHT ..... SEE TABLE I



SECTION B-B

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
10	STAPLE	FOR 1 1/4" STRAP	48	9	3	1 1/4
9	STIFFENER	2 x 6 x 74	32	-	-	-
8	SEAL	FOR 1 1/4" STRAP	32	-	-	-
7	STEEL STRAP	1 1/4 x .035 x 19FT 3	16	-	-	-
6	CHAIN & LOADBINDER	3/8	8	-	-	-
5	SLEEPER	2 x 4 x 18	20	SEE FIELD NOTE		
4	BRACE, CLEAT	2 x 6 x 36 1/2	8	TRAILER FLOOR	6	16d
3	BRACE, DIAGONAL	2 x 6 TO SUIT	16	1 & 4	4 PER JOINT	16d
2	BRACE, BACKUP	2 x 10 x 48	8	4	3	16d
1	BRACE, UPRIGHT	2 x 6 x 46 1/2	8	2	6	8d

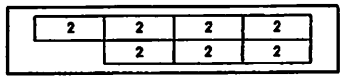
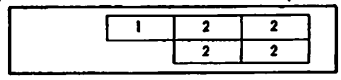
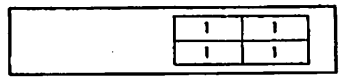
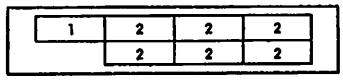
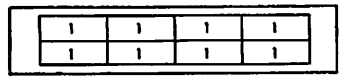
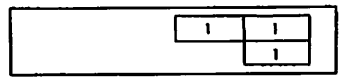
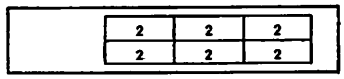
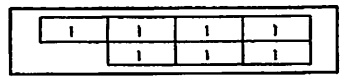
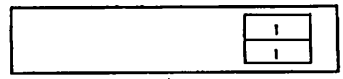
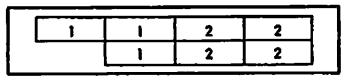
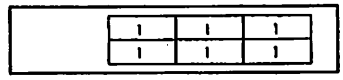
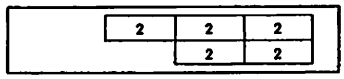
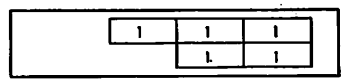
LIST OF MATERIALS AND NAILING DATA

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LTL 40 FT TRAILER (FLATBED)

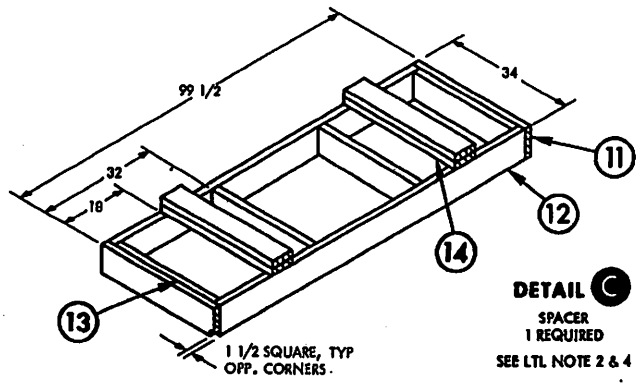
1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD QUANTITIES, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF UNIT LOADS TO BE SHIPPED FROM TABLE I AND POSITION THEM ON THE TRAILER ACCORDINGLY. SHIPMENT OF 15 UNIT LOADS IS NOT AUTHORIZED.

TABLE I

NO. OF UNIT LOADS	LOAD PATTERN* FWD	NO. OF UNIT LOADS	LOAD PATTERN* FWD	NO. OF UNIT LOADS	LOAD PATTERN* FWD
14		9		4	
13		8		3	
12		7		2	
11		6		1	
10		5			

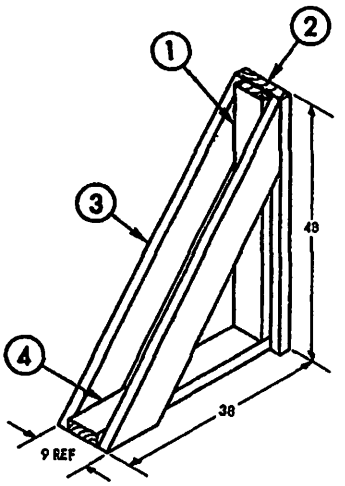
*LOAD PATTERN SHOWS THE FLOOR PLAN FOR THE LOADED TRAILER. THE NUMBER IN THE BLOCKS IS THE NUMBER OF UNIT LOADS IN THE LAYER.

2. WHEN SHIPPING LTL LOADS OF 9, 11 OR 13 UNIT LOADS (SEE TABLE I) WHERE A ONE HIGH UNIT LOAD(S) BACK UP A STACK OF TWO HIGH UNIT LOADS, A SPACER ASSEMBLY MUST BE SECURE TO ALL ONE HIGH LOADS. KNEE BRACES ARE USED TO PREVENT THE LONGITUDINAL MOVEMENT OF THE ONE HIGH UNIT LOADS AND THE SPACER ASSEMBLY. SEE LTL ON PAGE 5.
3. WHEN SHIPPING A LTL OF 8 OR LESS UNIT LOADS, THE FORE AND AFT BLOCKING SHALL BE ACCOMPLISHED AS SHOWN ON PAGE 6.
4. THE LENGTH OF THE SPACER ASSEMBLY, DETAIL C SHALL BE SUCH AS TO BEAR AGAINST THE UNIT LOAD IN FRONT OF IT AND THE KNEE BRACE AT THE REAR. SHIM TO FILL VOIDS IF PRESENT USING SUITABLE THICKNESS MATERIAL AND SECURE SHIM IN PLACE BY NAILING TO SPACER ASSEMBLY. SECURE SPACER TO UNIT LOAD TOP ADAPTER RAILS WITH 3/4 x .035 (4 PLACES). SEAL WITH ONE DOUBLE CRIMPED SEAL.
5. THE LTL ON PAGE 5 IS FOR NINE UNIT LOADS AND THE LTL ON PAGE 6 IS FOR FIVE UNIT LOADS. LTL SHIPMENTS ARE NOT LIMITED TO THIS QUANTITY. ACTUAL SHIPPING REQUIREMENTS WILL DETERMINE THE NUMBER OF LOADS TO BE SHIPPED AND THEY SHOULD BE BLOCKED, BRACED AND TIED DOWN IN ACCORDANCE WITH THE APPROPRIATE PARTS OF THIS STANDARD.



DETAIL C
SPACER
1 REQUIRED
SEE LTL NOTE 2 & 4

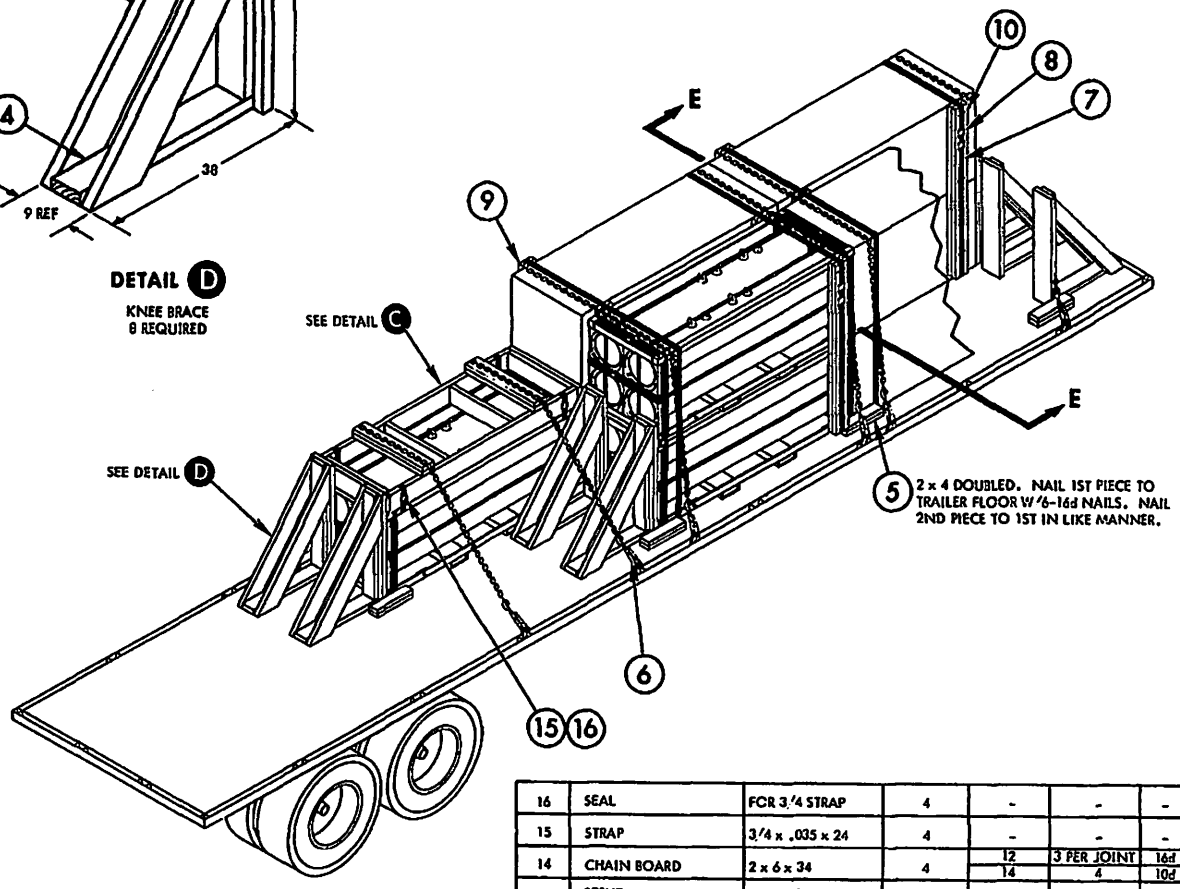
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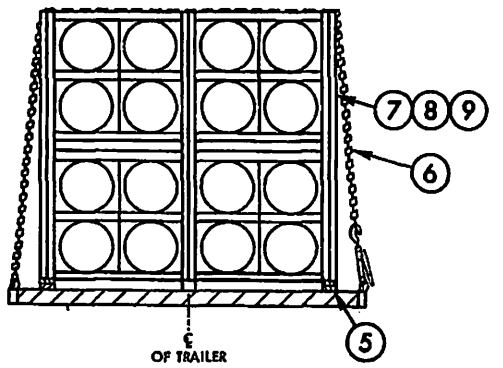
DETAIL D
KNEE BRACE
8 REQUIRED

WARNING
DO NOT POSITION CHAIN OVER STEEL STRAPPING.

POSITION FORWARD UNIT LOADS 41 INCHES FROM FRONT OF TRAILER.



5 2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

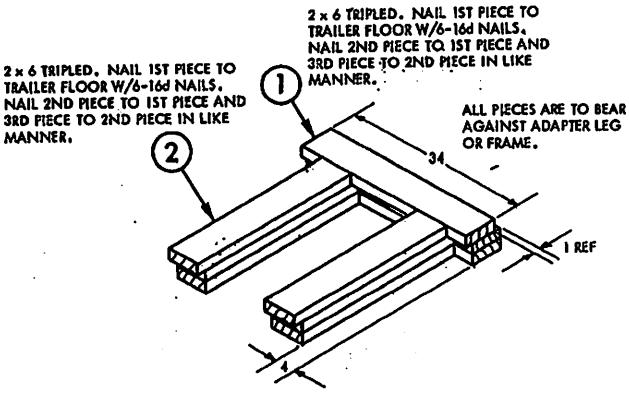


SECTION E-E

PIECE NO.	DESCRIPTION	SIZE	NO.PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
16	SEAL	FOR 3/4 STRAP	4	-	-	-
15	STRAP	3/4 x .035 x 24	4	-	-	-
14	CHAIN BOARD	2 x 6 x 34	4	12 14	3 PER JOINT 4	16d 10d
13	STRUT	2 x 8 x 31	4	SEE 12		
12	LONGITUDINAL	2 x 8 x 96 1/2	2	13	4 PER JOINT	16d
11	END	2 x 8 x 34	2	12	4 PER JOINT	16d
10	STAPLE	FOR 1 1/4 STRAP	24	-	-	-
9	STIFFENER	2 x 6 x 74	16	-	-	-
8	SEAL	FOR 1 1/4 STRAP	16	-	-	-
7	STEEL STRAP	1 1/4 x .035 x 19 FT 3	8	-	-	-
6	CHAIN & LOADBINDER	3/8	6	-	-	-
5	SLEEPER	2 x 4 x 18	18	SEE FIELD NOTE		
4	BRACE, CLEAT	2 x 6 x 36 1/2	8	TRAILER FLOOR	6	16d
3	BRACE, DIAGONAL	2 x 6 TO SUIT	16	1 & 4	4 PER JOINT	16d
2	BRACE, BACKUP	2 x 10 x 48	8	4	3	16d
1	BRACE, UPRIGHT	2 x 6 x 46 1/2	8	2	6	8d

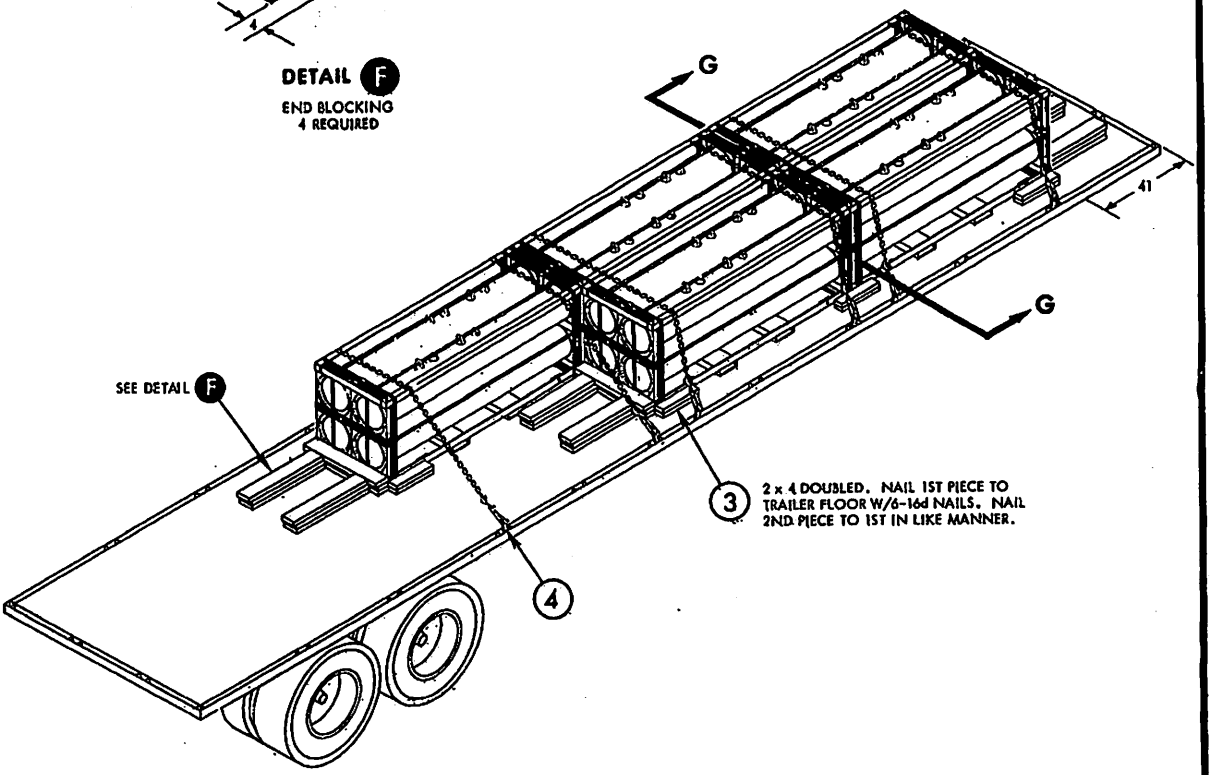
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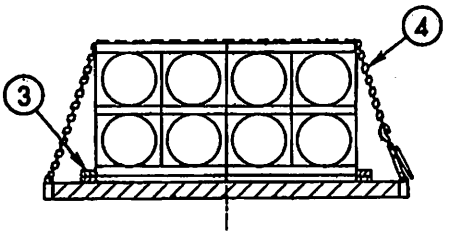
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DETAIL F
END BLOCKING
4 REQUIRED

WARNING
DO NOT POSITION CHAIN OVER STEEL STRAPPING.





SECTION G-G

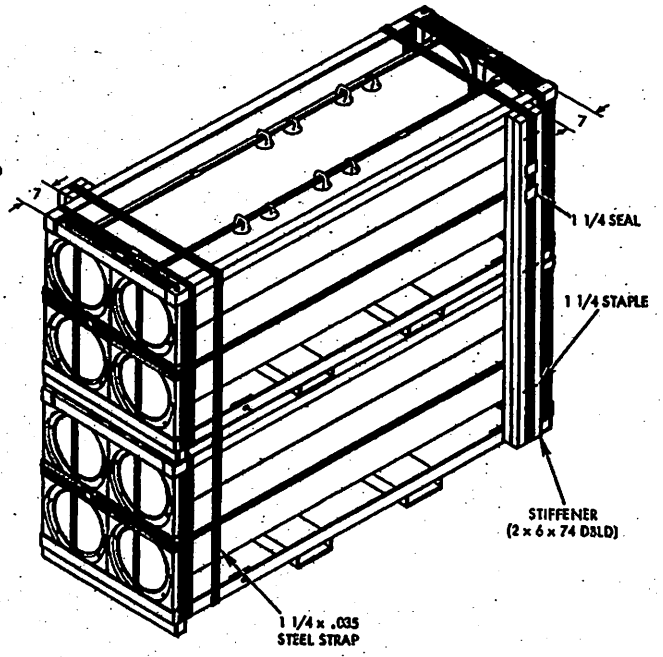
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
4	CHAIN & LOADBINDER	3/8	6	-	-	-
3	SLEEPER	2 x 4 x 18	16	SEE FIELD NOTE		
2	CLEAT	2 x 6 x 32	24	SEE DETAIL F		
1	CROSSMEMBER	2 x 6 x 34	12	SEE DETAIL F		

LIST OF MATERIALS AND NAILING DATA

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UNITIZING PROCEDURE

1. WHEN REQUIRED BY THE FTL OR LTL OF THIS DOCUMENT, THE TWO HIGH STACKED UNIT LOADS SHALL BE STRAPPED TOGETHER AS SHOWN.
2. THE DOUBLED 2 x 6 STIFFENERS SHALL EXTEND FROM THE TOP OF THE UPPER UNIT LOAD TO THE BOTTOM EDGE OF THE BASE FRAME OF THE LOWER UNIT LOAD.
3. THE 1 1/4 STRAPS POSITIONED AS SHOWN ENCIRCLE THE STACKED UNIT LOADS PASSING UNDER THE BOTTOM FRAME OF THE LOWER UNIT LOAD HOLDING THE STIFFENERS IN PLACE.
4. TENSION STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS OR ONE DOUBLE NOTCHED SEAL. SECURE EACH STRAP TO THE STIFFENER WITH THREE 1 1/4 INCH STAPLES.



REVIEW ACTIVITIES:
NAVY - OS, AS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N241)

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